🤖✨ Responsible use of AI tools

How to use and report the use of AI tools

Gen AI tools (ChatGPT, Gemini, etc.) can be excellent assistants for generating ideas and debugging code, but they are not a replacement for your own critical thinking.

Guiding Principles for this Class:

- **Verify Everything:** AI makes mistakes. You are responsible for the accuracy of your submission.
- **Active vs. Passive:** Engage in **Active Learning** (using AI to understand the 'how') rather than **Passive Reception** (copy-pasting the result). Use AI to help you build the ladder, not to climb it for you.
- **Transparency:** If you use AI, you must append a list of the prompts you used to the end of your assignment.

The example below provides a clear template for what a responsible submission looks like. For more tips on citations for Gen AI tools usage, you can visit **this (https://engineering.oregonstate.edu/tools-services/academic-support/coe-academic-integrity/citation-guidelines-code-ai-sources#:~:text=This%20transparency%20is%20crucial%20for,Retrieved%20from%20%5BURL%5D.)** page.

Example Scenario: The "Paired t-test" Homework

The Assignment:

Dr. Smith wants to test if a new study technique improves quiz scores. She collects scores from 10 students before and after they use the technique. Perform a paired t-test in R to determine if there is a significant effect of the new study technique.

1. ❌ **The "Wrong" Way (Academic Dishonesty)**

Student Action: The student copies the entire prompt into ChatGPT and asks it to solve the problem for them.

- **Prompt:** "Here are the numbers: [list]. Write code to do a paired t-test and tell me if it is significant and what the p-value means."
- **Result:** The AI writes the code, runs the numbers, interprets the p-value, and writes the conclusion paragraph.

- **Submission:** The student pastes the AI's code and conclusion directly into their homework without understanding it.
- **Verdict:** This is plagiarism/contract cheating. The student demonstrated zero learning.

2. ✅ **The "Right" Way (AI as a Tutor/Assistant)**

Student Action: The student knows they need a paired t-test but forgets the specific syntax for the function or how to format the data.

Step A: The Prompt

Student Prompts AI:

"I need to run a paired t-test in R. I have two lists of scores, 'pre' and 'post'. Can you show me the syntax for the t.test function for paired data? Don't solve it for me, just show me the code structure."

Step B: The AI Response

AI Output:

"Sure! In R, you can use the t.test() function with the argument paired = TRUE. Here is the general syntax:

t.test(post_scores, pre_scores, paired = TRUE)

Make sure both vectors are the same length!"

Step C: The Student Work

The student takes this syntax, writes their *own* code using their actual data variables, adds comments explaining what they are doing, and writes their own conclusion based on the output.

3. 📄 **What the Student Submits**

If a student uses Gen AI in the "Right" way, their submission should look like this. It includes **(A) The Code** and **(B) The AI Disclosure**.

Part A: The Code (Annotated)

R

```
# Data entry (Entered by me)

pre_scores <- c(75, 80, 65, 70, 88, 92, 76, 82, 79, 85)

post_scores <- c(80, 85, 70, 72, 90, 95, 78, 85, 82, 88)
```

```
# Performing the Paired t-test

# I used AI to verify the correct flag is 'paired = TRUE' rather than 'type = paired'

result <- t.test(post_scores, pre_scores, paired = TRUE)


# Outputting the result

print(result)
```

Part B: The AI Disclosure Statement (Required)

Place this at the bottom of your assignment.

Generative AI Disclosure

- **Tool Used:** ChatGPT (OpenAI, GPT-4o)[1]
- **Purpose:** I used the tool to recall the correct syntax for the t-test function in R.
- **Prompt Used:** "Show me the syntax for a paired t-test in R using two vectors. Do not solve a problem, just show the function."
- **Modifications:** The AI provided generic variables (x and y). I adapted this to use my specific variable names (pre_scores and post_scores) and wrote the data entry lines myself. I interpreted the p-value in the conclusion paragraph entirely on my own.

Summary Checklist for Students

Before submitting, ask yourself:

1. **Did I write the code?** (Or did I just Paste?)
2. **Did I write the interpretation?** (The "Why" and "What it means" must come from you.)
3. **Did I disclose?** (Include the standard disclosure statement).